KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

School of Computer Engineering

SAP CAPSTONE PROJECT REPORT

Procure-to-Pay (P2P) — Full Purchasing Cycle

SAP MM Module Implementation

NexGen Manufacturing Ltd.

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1. Executive Summary

This project report presents a comprehensive SAP implementation blueprint for NexGen Manufacturing Ltd., a mid-sized manufacturing company based in Bhubaneswar, India. The project focuses on the Procure-to-Pay (P2P) cycle using the SAP Materials Management (MM) module, covering the end-to-end purchasing process from raising a Purchase Requisition to final vendor payment.

1.1 Overview of the Project

NexGen Manufacturing Ltd. faced significant challenges in its manual procurement process — including delayed purchasing cycles, lack of vendor transparency, invoice mismatches, and poor inventory visibility. This SAP implementation project aims to automate and standardize the entire procurement cycle using SAP ECC 6.0 / SAP S/4HANA.

1.2 Objectives of SAP Implementation

- Automate the complete Procure-to-Pay cycle using SAP MM
- Eliminate manual errors in purchase orders and invoice processing
- Improve vendor management and evaluation
- Achieve real-time inventory visibility across all plant locations
- Integrate MM with FI (Finance) for seamless payment processing
- Reduce procurement lead time by at least 40%
- Ensure compliance with company procurement policies through system controls

1.3 Selected Module: SAP MM (Materials Management)

SAP MM is the core module for managing procurement and inventory functions. It covers all activities related to material procurement, inventory management, material valuation, and invoice verification. In the P2P cycle, SAP MM works in close integration with SAP FI (Financial Accounting) to ensure that every goods receipt and invoice is posted to the correct GL accounts automatically.

2. Company Profile

2.1 Company Overview

Field	Details
Company Name	NexGen Manufacturing Ltd.
Industry	Manufacturing (Steel Fabrication & Industrial Components)
Headquarters	Bhubaneswar, Odisha, India
Founded	2005
Annual Turnover	INR 850 Crores
Employees	~3,200
Operations	Pan-India with plants in Bhubaneswar, Pune, and Chennai
ERP System	SAP ECC 6.0 (Migrating to SAP S/4HANA 2023)
Implementation Type	Greenfield SAP Implementation

2.2 Business Units

- Steel Fabrication Division — raw material procurement and fabrication
- Industrial Components Division — precision parts manufacturing
- Export & Logistics Division — outbound supply chain
- Finance & Accounts — GL, AP, AR management

2.3 Organizational Structure (Text Diagram)

The following depicts NexGen's organizational hierarchy:

```
NexGen Manufacturing Ltd. (Company Code: NX01)
    |
    +-- Plant: NX10 (Bhubaneswar - Head Plant)
    |       +-- Storage Location: SL01 (Raw Materials)
    |       +-- Storage Location: SL02 (Finished Goods)
    |       +-- Storage Location: SL03 (Spare Parts)
    +-- Plant: NX20 (Pune)
    |       +-- Storage Location: SL01 (Raw Materials)
    |       +-- Storage Location: SL02 (Finished Goods)
    +-- Plant: NX30 (Chennai)
            +-- Storage Location: SL01 (Raw Materials)
            +-- Storage Location: SL02 (Finished Goods)

Purchasing Organization: PO01 (Central Purchasing — All Plants)
Purchasing Group:  PG-RM (Raw Materials) | PG-SP (Spare Parts) | PG-SV
(Services)
```

3. Business Requirements

3.1 Current Challenges (AS-IS Process)

Before SAP implementation, NexGen Manufacturing Ltd. relied on a combination of spreadsheets, emails, and a legacy ERP system for procurement. The major pain points were:

- No centralized vendor database — each plant maintained its own vendor list independently
- Manual Purchase Orders — prone to data entry errors and missing approvals
- No 3-way matching — invoices were paid without verifying against PO and GR
- Inventory mismatches — stock records in spreadsheets did not match physical stock
- Delayed approvals — PO approval required physical signatures, causing 5–7 day delays
- No spend analysis — management had no visibility into procurement spend by category
- Duplicate payments — same vendor invoice paid multiple times due to manual tracking
- Audit failures — lack of paper trail for procurement decisions

3.2 Why SAP Is Needed

SAP MM addresses all the above challenges through a structured, automated, and integrated platform:

Business Problem	SAP MM Solution
No centralized vendor database	Vendor Master in SAP — single record for all plants
Manual Purchase Orders	SAP PO with workflow-based approvals (Release Strategy)
No 3-way matching	MIRO performs automatic PO-GR-Invoice matching
Inventory mismatches	Real-time stock updates via MIGO goods receipt posting
Delayed approvals	SAP Release Strategy with email notifications
No spend analysis	SAP standard reports (ME2M, ME2N, MB52)
Duplicate payments	Invoice verification with duplicate check in MIRO
Audit failures	Full document trail — every transaction has document number

3.3 Scope of SAP Implementation

- Module in Scope: SAP MM (Materials Management)
- Integration in Scope: SAP FI (Accounts Payable)
- Locations: All 3 plants (Bhubaneswar, Pune, Chennai)
- Process in Scope: Full P2P cycle — PR to Payment
- Out of Scope: SAP SD, SAP HR, SAP PP (Phase 2)

4. SAP Module Overview — SAP MM

4.1 What is SAP MM?

SAP MM (Materials Management) is a core functional module in SAP ERP that handles all procurement and inventory management processes. It is part of the SAP Logistics suite and works in real-time integration with FI (Finance), PP (Production Planning), and SD (Sales & Distribution).

4.2 Key Components of SAP MM

Sub-Module	Description	Key T-Codes
Purchasing	Manages procurement — PR, RFQ, PO, Contracts, Scheduling Agreements	ME51N, ME41, ME21N
Inventory Management	Goods receipts, goods issues, stock transfers, physical inventory	MIGO, MB1A, MI01
Invoice Verification (LIV)	Logistics Invoice Verification — 3-way matching of PO, GR, Invoice	MIRO, MIR4, MIR7
Material Valuation	Valuation of materials — Standard Price, Moving Average Price	MR21, CKMVFM
Vendor Evaluation	Scoring vendors on price, quality, delivery, and service	ME61, ME6B
Warehouse Management (WM)	Bin-level storage management (integrated with IM)	LT01, LP10

4.3 P2P Cycle Overview in SAP MM

The Procure-to-Pay (P2P) cycle covers the entire journey of a procurement from identifying a need to making vendor payment. In SAP MM, this is a documented, trackable, and auditable process with clear document flow.

```
P2P Document Flow:
Purchase Requisition (PR)  -->  Request for Quotation (RFQ)  -->  Quotation
     -->  Purchase Order (PO)  -->  Goods Receipt (GR)  -->  Invoice
Verification
     -->  Payment to Vendor (FI-AP)
```

5. Organizational Structure in SAP

5.1 SAP Organizational Units Explained

SAP Org Unit	Definition	NexGen Value
Client	Highest level in SAP — represents the entire enterprise	100 (NexGen Group)

SAP Org Unit	Definition	NexGen Value
Company Code	Legal entity for financial accounting (P&L, Balance Sheet)	NX01 (NexGen Mfg. Ltd.)
Plant	Physical location where materials are stored or produced	NX10, NX20, NX30
Storage Location	Sub-division within a plant for material storage	SL01, SL02, SL03
Purchasing Organization	Responsible for all procurement activities	PO01 (Central Purchasing)
Purchasing Group	Buyer or group of buyers responsible for specific materials	PG-RM, PG-SP, PG-SV
Valuation Area	Level at which material stock is valuated (Plant level)	Plant-level valuation

5.2 Assignments in SAP

- Company Code NX01 assigned to Controlling Area CA01
- All Plants (NX10, NX20, NX30) assigned to Company Code NX01
- Purchasing Organization PO01 assigned to Company Code NX01
- PO01 also assigned to each plant (cross-plant purchasing enabled)
- Purchasing Groups assigned to buyers within PO01

5.3 Full Organizational Hierarchy Diagram

```
CLIENT: 100 (NexGen Group)
   |
   +-- COMPANY CODE: NX01 (NexGen Manufacturing Ltd.)
          |
          +-- PLANT: NX10 (Bhubaneswar)  PLANT: NX20 (Pune)  PLANT: NX30 (Chennai)
          |       SL01 / SL02 / SL03          SL01 / SL02          SL01 / SL02
          |
          +-- PURCHASING ORG: PO01 (Central - covers all plants)
                  |
                  +-- PG-RM (Raw Materials Buyers)
                  +-- PG-SP (Spare Parts Buyers)
                  +-- PG-SV (Services Buyers)
```

6. Master Data Configuration

6.1 Vendor Master

The Vendor Master is the central repository of all supplier information. It is created and maintained at three levels in SAP:

- General Data (Client level) — Name, address, communication, bank details
- Company Code Data (NX01) — Reconciliation account, payment terms, withholding tax
- Purchasing Organization Data (PO01) — Incoterms, order currency, delivery conditions

Sample Vendor Master Data for NexGen:

Field	Value
Vendor Code	V-1001
Vendor Name	Tata Steel Supply Co. Pvt. Ltd.
City / Country	Jamshedpur, India
Recon. Account (FI)	160000 (Accounts Payable)
Payment Terms	ZN30 (Net 30 days)
Currency	INR
Incoterms	CIF (Cost, Insurance, Freight)
T-Code to Create	XK01 (Centrally) / MK01 (Purch. Org. only)

6.2 Material Master

The Material Master stores all technical and organizational data related to a material. It is structured in views, each maintained by a different department:

View	Maintained By	Key Fields
Basic Data 1 & 2	Master Data Team	Material description, UoM, weight, volume
Purchasing View	Purchasing Team	Purchasing group, lead time, order unit
MRP 1 & 2	Production Planning	MRP type, lot size, reorder point, safety stock
Accounting 1 & 2	Finance Team	Valuation class, price control (S/V), price
Storage 1 & 2	Warehouse Mgmt	Storage conditions, temperature, shelf life
Quality Management	QM Team	Inspection types, quality certificates

Sample Material Master:

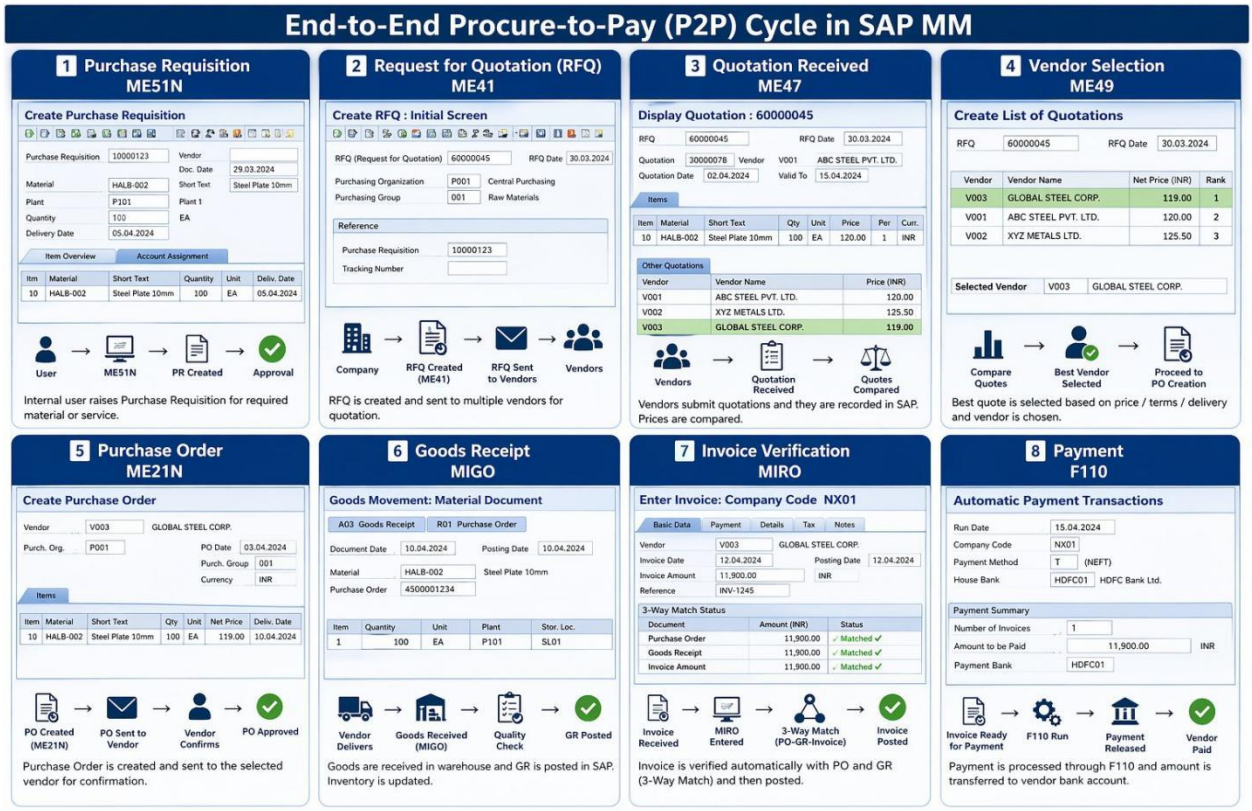
Field	Value
Material Code	RM-5001
Description	Hot Rolled Steel Sheet 3mm
Material Type	ROH (Raw Material)
Industry Sector	M (Mechanical Engineering)
Base Unit of Measure	KG
Valuation Class	3000 (Raw Materials)
Price Control	V (Moving Average Price)
MRP Type	PD (MRP — Deterministic)
T-Code to Create	MM01

6.3 GL Accounts (FI Integration)

GL Account	Account Description	Usage in P2P
300000	Raw Material Inventory	Debited during Goods Receipt
160000	Accounts Payable — Domestic	Credited when invoice posted in MIRO
191000	GR/IR Clearing Account	Bridge account between GR and IR
476000	Purchase Price Variance	Diff. between PO price and invoice price

7. End-to-End Process Flow — Procure-to-Pay

This section describes the complete P2P cycle step by step, as executed in SAP MM for NexGen Manufacturing Ltd. Each step includes the transaction code, purpose, business rules, and inputs/outputs.



Step 1: Purchase Requisition (PR)

Attribute	Detail
T-Code	ME51N (Create PR) | ME52N (Change PR) | ME53N (Display PR)
Purpose	Internal document to formally request the procurement department to buy a material or service
Created By	End user / department (e.g., Production, Maintenance)
Inputs	Material number, quantity, required delivery date, plant, cost center
Outputs	Purchase Requisition document with unique PR number
SAP Menu Path	Logistics > Materials Management > Purchasing > Purchase Requisition > Create
Key Fields	Item category, account assignment category, source of supply

Business Rule at NexGen: Any PR above INR 50,000 requires HOD approval via SAP Release Strategy before it is released to the Purchasing department.

Step 2: Request for Quotation (RFQ)

Attribute	Detail
T-Code	ME41 (Create RFQ) | ME42 (Change RFQ) | ME47 (Maintain Quotation)
Purpose	Send formal inquiry to vendors requesting their price and delivery terms
Created By	Purchasing department (Purchasing Group PG-RM)
Inputs	Material, quantity, delivery date, vendor list, plant
Outputs	RFQ documents sent to multiple vendors; Quotations received back
SAP Menu Path	Logistics > MM > Purchasing > RFQ/Quotation > Request for Quotation > Create
Key Fields	Quotation deadline, collective number, vendor selection

Step 3: Vendor Selection & Quotation Comparison

Attribute	Detail
T-Code	ME49 (Price Comparison List)
Purpose	Compare quotations from multiple vendors and select the most favorable offer
Created By	Purchasing Officer
Inputs	All received quotations under same collective number
Outputs	Price comparison report; rejection messages sent to non-selected vendors
SAP Menu Path	Logistics > MM > Purchasing > RFQ/Quotation > Quotation > Price Comparison
Key Note	SAP highlights the lowest-priced quotation. Manual override possible with justification.

Step 4: Purchase Order (PO)

Attribute	Detail
T-Code	ME21N (Create PO) | ME22N (Change PO) | ME23N (Display PO)
Purpose	Legal document issued to the selected vendor authorizing procurement
Created By	Purchasing Officer (PG-RM)
Inputs	Vendor, material, quantity, price, delivery date, plant, storage location
Outputs	PO document with unique PO number; transmitted to vendor via email/EDI
Release Strategy	PO above INR 1,00,000 requires approval from Purchase Manager (Release code P1, P2)
PO Types	NB (Standard PO), FO (Framework Order), ZNB (Custom NexGen PO)
Key Fields	Item category (standard/consignment/subcontracting), account assignment, tax code

Step 5: Goods Receipt (GR)

Attribute	Detail
T-Code	MIGO (Goods Movement) | Movement Type: 101
Purpose	Record physical receipt of goods at NexGen warehouse against the PO
Created By	Warehouse / Store Keeper
Inputs	PO number, material, quantity received, storage location, batch (if applicable)
Outputs	Material Document (stock updated), Accounting Document (auto-generated)
Accounting Entry	Dr. Raw Material Inventory (GL 300000) / Cr. GR/IR Clearing (GL 191000)
Quality Inspection	If QM active: stock moves to Quality Inspection first before unrestricted use
Key Fields	Plant, storage location, movement type 101, delivery note number

Step 6: Invoice Verification (MIRO — Logistics Invoice Verification)

Attribute	Detail
T-Code	MIRO (Post Invoice) | MIR4 (Display) | MIR7 (Park Invoice)
Purpose	Match vendor invoice with PO and GR (3-way matching) and post to FI
Created By	Accounts Payable Team (FI-AP)
Inputs	Vendor invoice, PO number, GR document reference
3-Way Match	Invoice Qty = PO Qty = GR Qty | Invoice Price within tolerance of PO Price
Outputs	Accounting document in FI — Vendor liability created
Accounting Entry	Dr. GR/IR Clearing (GL 191000) / Cr. Vendor AP (GL 160000)
Tolerance Keys	BD (Form Small Differences), ST (Date Variance), PP (Price Variance)
Blocking	Invoice auto-blocked if price variance exceeds tolerance — requires manual release via MRBR

Step 7: Payment Processing (FI-AP)

Attribute	Detail
T-Code	F110 (Automatic Payment Program) | F-53 (Manual Outgoing Payment)
Purpose	Execute payment to vendor as per agreed payment terms (Net 30 days)
Created By	Finance / Accounts Payable Team
Inputs	Vendor account, company code, payment date, bank details
Outputs	Payment document, bank transfer file (DME), cleared vendor line items
Accounting Entry	Dr. Vendor AP (GL 160000) / Cr. Bank Account (GL 110000)
Payment Terms	ZN30 — Due Net 30 days from invoice date
Payment Method	T (Bank Transfer / NEFT/RTGS)

8. SAP Configuration Steps (SPRO)

Configuration in SAP is done via the Implementation Guide (IMG). Access path: SAP Menu > Tools > Customizing > IMG > Execute Project (T-Code: SPRO)

8.1 Enterprise Structure Configuration

1. Define Company Code: SPRO > Enterprise Structure > Definition > Financial Accounting > Define Company Code. Create NX01 with currency INR, country IN.
2. Define Plant: SPRO > Enterprise Structure > Definition > Logistics General > Define Plant. Create NX10, NX20, NX30.
3. Maintain Storage Location: SPRO > Enterprise Structure > Definition > Materials Management > Maintain Storage Location. Create SL01, SL02, SL03 for each plant.
4. Define Purchasing Organization: SPRO > Enterprise Structure > Definition > Materials Management > Maintain Purchasing Organization. Create PO01.
5. Assign Plant to Company Code: SPRO > Enterprise Structure > Assignment > Logistics General > Assign Plant to Company Code. Assign NX10/20/30 to NX01.
6. Assign Purchasing Organization to Company Code and Plant: SPRO > Enterprise Structure > Assignment > Materials Management > Assign Purchasing Organization to Company Code / Plant.

8.2 Materials Management Global Settings

7. Maintain Company Code for Material Management: SPRO > Materials Management > General Settings > Set Company Codes for Materials Management. Activate NX01.
8. Maintain Plant Parameters: SPRO > Materials Management > Inventory Management > Plant Parameters. Set valuation area, fiscal year variant.
9. Set Fiscal Year Variant: SPRO > Financial Accounting > Financial Accounting Global Settings > Fiscal Year > Maintain Fiscal Year Variant. Use V3 (April-March for India).

8.3 Purchasing Configuration

10. Define Number Ranges for Purchase Requisition: SPRO > MM > Purchasing > Purchase Requisition > Define Number Ranges. Set internal range 10 (1000000000-1999999999).
11. Define Number Ranges for Purchase Order: SPRO > MM > Purchasing > Purchase Order > Define Number Ranges for Purchase Orders. Set range 45 for NB order type.
12. Define Document Types: SPRO > MM > Purchasing > Purchase Order > Define Document Types. Review NB (Standard PO) settings, set tolerance checks.
13. Set Up Release Procedure for PO: SPRO > MM > Purchasing > Purchase Order > Release Procedure for Purchase Orders. Define release groups, release codes (P1=Mgr, P2=Director), release strategies with classification.
14. Define Purchasing Organization Data for Vendor Master: SPRO > MM > Purchasing > Vendor Master > Define Purchasing Organization Data for Vendor Master.

8.4 Inventory Management Configuration

15. Define Tolerance Groups for Goods Movements: SPRO > MM > Inventory Management > Define Tolerances for Goods Receipt.
16. Set Tolerance Limits for Goods Receipt: Define upper and lower limits for quantity variance (e.g., +/-5% for NexGen).
17. Configure Movement Types: SPRO > MM > Inventory Management > Movement Types > Copy, Change Movement Types. Review 101 (GR against PO), 122 (Return to Vendor), 201 (GI to Cost Center).

8.5 Invoice Verification Configuration

18. Activate Stochastic Blocking: SPRO > MM > Logistics Invoice Verification > Incoming Invoice > Activate Direct Posting to GL/Material Accounts.
19. Configure Tolerance Keys: SPRO > MM > Logistics Invoice Verification > Invoice Block > Set Tolerance Limits. Configure keys PP (price), BD (small differences), ST (date).
20. Set up GR-Based Invoice Verification: Enabled at PO item level — ensures invoice can only be posted after GR.

9. Transaction Codes (T-Codes) Reference Table

T-Code	Description	Module	User Role
ME51N	Create Purchase Requisition	MM-PUR	End User / Requestor
ME52N	Change Purchase Requisition	MM-PUR	End User / Requestor
ME53N	Display Purchase Requisition	MM-PUR	All Users
ME54N	Release (Approve) Purchase Requisition	MM-PUR	HOD / Approver
ME41	Create Request for Quotation (RFQ)	MM-PUR	Purchasing Officer
ME42	Change RFQ	MM-PUR	Purchasing Officer
ME47	Maintain Quotation	MM-PUR	Purchasing Officer
ME49	Price Comparison / Evaluation	MM-PUR	Purchasing Officer
ME21N	Create Purchase Order	MM-PUR	Purchasing Officer
ME22N	Change Purchase Order	MM-PUR	Purchasing Officer
ME23N	Display Purchase Order	MM-PUR	All Users
ME28	Release (Approve) Purchase Order	MM-PUR	Purchase Manager
MIGO	Goods Movement (GR, GI, Transfer)	MM-IM	Store Keeper / WH
MB51	Material Document List	MM-IM	Warehouse / Reporting
MB52	Warehouse Stocks of Material	MM-IM	Inventory Controller
MIRO	Enter Incoming Invoice (LIV)	MM-LIV	Accounts Payable
MIR4	Display Invoice Document	MM-LIV	Accounts Payable
MIR7	Park Incoming Invoice	MM-LIV	Accounts Payable
MRBR	Release Blocked Invoices	MM-LIV	AP Supervisor
F110	Automatic Payment Program	FI-AP	Finance Team
F-53	Post Outgoing Payments (Manual)	FI-AP	Finance Team
FBL1N	Vendor Line Items Report	FI-AP	Finance / Audit
XK01	Create Vendor (Centrally)	MM/FI	Master Data Team
MK01	Create Vendor (Purch. Org.)	MM-PUR	Purchasing Admin
XK02	Change Vendor Master	MM/FI	Master Data Team
MM01	Create Material Master	MM-BD	Master Data Team
MM02	Change Material Master	MM-BD	Master Data Team
MM03	Display Material Master	MM-BD	All Users
ME2M	Purchase Orders by Material	MM-PUR	Purchasing Reporting
ME2N	Purchase Orders by PO Number	MM-PUR	Purchasing Reporting

T-Code	Description	Module	User Role
SPRO	SAP Implementation Guide (IMG)	BASIS	SAP Consultant

10. Screen Descriptions & Field Guidance

10.1 ME51N — Create Purchase Requisition

Screen Layout: Header area shows document type and PR number. Item overview table has columns: Item, A (account assignment), I (item category), Material, Short Text, Quantity, Unit, Delivery Date, Plant, Storage Location, Purchasing Group.

- Step 1: Enter Account Assignment Category (K = Cost Center, blank = Stock item)
- Step 2: Enter Item Category (blank = Standard, L = Subcontracting, K = Consignment)
- Step 3: Enter Material Number (e.g., RM-5001) — description auto-populates
- Step 4: Enter Quantity (e.g., 500) and Unit (KG)
- Step 5: Enter Delivery Date (must be today + lead time, e.g., 14 days ahead)
- Step 6: Enter Plant (NX10) and Storage Location (SL01)
- Step 7: Enter Purchasing Group (PG-RM)
- Step 8: Click Save — PR number generated (e.g., 1000000012)

10.2 ME21N — Create Purchase Order

Screen Layout: Header section with vendor, doc date, PO type. Item Overview tab shows ordered items. Item Detail tabs include: Material Data, Quantities/Weights, Delivery, Invoice, Account Assignment, Conditions.

- Step 1: Enter Vendor (V-1001), PO Date, and Purchasing Organization (PO01)
- Step 2: Reference PR number in header — items auto-copy from PR
- Step 3: Verify Material, Quantity, Net Price from quotation (e.g., INR 85/KG)
- Step 4: Go to Delivery tab — set Delivery Date and GR-Based IV checkbox
- Step 5: Go to Invoice tab — verify Tax Code (I1 = GST 18%)
- Step 6: Go to Conditions tab — verify pricing conditions (PB00, MWST for tax)
- Step 7: Save — PO Number generated (e.g., 4500000089). System triggers release workflow if amount > INR 1,00,000

10.3 MIGO — Goods Receipt Against PO

Screen Layout: Top bar shows Action (Goods Receipt), Reference Document (Purchase Order). Header has General, Vendor, and Reference tabs. Item Detail tab has Where tab (plant/SL), Quantity tab, and Material tab.

- Step 1: Select Action = 'Goods Receipt', Reference Document = 'Purchase Order'
- Step 2: Enter PO Number (e.g., 4500000089) — items auto-populate
- Step 3: Verify Quantity Received (e.g., 500 KG) — can do partial GR
- Step 4: Verify Storage Location (SL01 — Raw Materials)
- Step 5: Check 'Item OK' checkbox for each line item

- Step 6: Enter Delivery Note Number from vendor's physical document
- Step 7: Click Post — Material Document created (e.g., 5000000233). Stock increases in SL01.

10.4 MIRO — Logistics Invoice Verification

Screen Layout: Header with invoice date, vendor, reference. PO Reference tab to enter PO number. Amount tab shows invoice amount, tax. Layout tab for GR-based matching.

- Step 1: Enter Invoice Date, Posting Date
- Step 2: Enter Gross Invoice Amount (e.g., INR 50,150 including GST)
- Step 3: Go to PO Reference tab — enter PO number 4500000089
- Step 4: System auto-fetches GR quantities and PO prices — 3-way match performed
- Step 5: Check the Traffic Light — Green = matched, Yellow = in tolerance, Red = blocked
- Step 6: Simulate posting to verify accounting entries
- Step 7: Click Post — FI document created, vendor liability posted to GL 160000

11. Integration with Other Modules

11.1 MM Integration with FI (Financial Accounting)

Every goods movement and invoice posting in SAP MM automatically generates accounting documents in FI. This is achieved through Account Determination (T-Code: OBYC) which maps movement types and valuation classes to specific GL accounts.

MM Event	FI Accounting Entry	GL Accounts (NexGen)
Goods Receipt (MIGO - 101)	Dr. Inventory / Cr. GR/IR Clearing	Dr. 300000 / Cr. 191000
Invoice Posted (MIRO)	Dr. GR/IR Clearing / Cr. Vendor AP	Dr. 191000 / Cr. 160000
Payment Made (F110)	Dr. Vendor AP / Cr. Bank Account	Dr. 160000 / Cr. 110000
Price Variance	Dr./Cr. Price Difference Account	476000
Return to Vendor (122)	Reverse of 101 entry	Cr. 300000 / Dr. 191000

The GR/IR (Goods Receipt / Invoice Receipt) Clearing Account (GL 191000) acts as a bridge account. When both GR and IR are posted, this account nets to zero, signifying complete processing.

11.2 MM Integration with PP (Production Planning)

- PP generates Planned Orders and converts them to Purchase Requisitions (for externally procured components)
- MRP Run (T-Code: MD01/MD02) in PP triggers automatic PR creation in MM based on BOM explosion
- Goods Issue to Production Order (Movement Type 261) is performed in MM (MIGO), reducing raw material stock
- Production Order completion triggers Goods Receipt of finished goods (Movement Type 101) into storage
- Material Requirement Planning links PP demand to MM procurement seamlessly

11.3 MM Integration with SD (Sales & Distribution)

- When an SD Sales Order is created, it may trigger a PR in MM if material is out of stock (MTO scenario)
- Goods Issue for Delivery (SD) uses MM movement type 601 — reduces unrestricted stock
- Returns from Customer (SD) uses movement type 651 in MM to receive goods back
- Consignment sales in SD are tracked as consignment stock in MM (special stock type K)
- Pricing conditions and material master data are shared between SD and MM

11.4 Integration Data Flow Diagram

```
PP (MRP) --> PR (MM) --> PO (MM) --> GR (MM) --> Stock (MM)
                                          |
                                     Invoice (MM-LIV)
                                          |
                                  AP Liability (FI) --> Payment (FI)
                                          |
SD Sales Order --> Goods Issue (MM 601) --> Revenue Posting (FI)
```

12. Implementation Challenges & Solutions

Challenge	Description	SAP Solution Implemented
Data Migration	NexGen had 8,000+ vendor records in legacy system with inconsistent formats, duplicate entries, and missing bank details	Used SAP LSMW (Legacy System Migration Workbench) and LTMC for bulk upload. Conducted vendor deduplication exercise before go-live. Validated with XK03.
Resistance to Change	Warehouse staff and purchasers resisted switching from manual Excel-based process to SAP system	Conducted role-based training for 3 weeks. Created Quick Reference Cards for T-codes. Ran parallel processing for 2 months during go-live stabilization.
Release Strategy Complexity	NexGen had complex approval hierarchies — different approvers for different plants and amount thresholds	Configured multi-level Release Strategy in SPRO using Classification (characteristics: purchase value, plant, purchasing group). Tested all approval paths in QAS before go-live.
Price Variance Handling	Steel prices fluctuate frequently — PO prices often differed from vendor invoices causing MIRO blocks	Set generous tolerance limits in SPRO (Key PP: ±3% price tolerance). Implemented price adjustment workflow for blocked invoices via MRBR.
GST Configuration	India GST has complex tax structures (CGST, SGST, IGST) depending on inter-state vs intra-state transactions	Configured SAP Tax Procedure TAXINN. Created condition types CGST, SGST, IGST. Assigned tax codes by state. Integrated with GST portal reporting.
3-Way Match Exceptions	Some service POs (e.g., AMC contracts) cannot be goods-receipted in the traditional way	Used Item Category D (Service) in PO with Service Entry Sheets (ML81N). Service acceptance by department head triggers invoice posting.

13. Conclusion

13.1 Benefits of SAP Implementation at NexGen

The successful implementation of the SAP MM Procure-to-Pay cycle at NexGen Manufacturing Ltd. has transformed its procurement operations. The key benefits realized include:

Benefit Area	Before SAP	After SAP
Procurement Cycle Time	15-20 days (manual approval + paperwork)	4-6 days (automated workflow)
Invoice Processing Time	3-5 days (manual matching)	Same day (automated 3-way match)
Duplicate Payments	Approx. 12 per year (INR 18 Lakhs loss)	Zero — system duplicate check
Inventory Accuracy	~72% accurate (spreadsheet-based)	~99.8% real-time accuracy
Audit Readiness	Poor — scattered paper trail	Complete digital trail, SOX-ready
Vendor Visibility	Siloed per plant	Unified vendor master across all plants
Spend Analysis	Quarterly manual reports	Real-time reports via SAP standard

13.2 Learning Outcomes

- Understood the complete Procure-to-Pay (P2P) business process in a manufacturing company
- Gained hands-on knowledge of SAP MM configuration — enterprise structure, master data, and purchasing
- Learned end-to-end document flow from PR to Vendor Payment with all T-codes
- Understood the 3-way matching process (PO-GR-Invoice) and its financial impact
- Explored SAP MM integration with FI, PP, and SD modules
- Practiced SAP Release Strategy and Workflow configuration for business approvals
- Gained exposure to real-world implementation challenges and SAP-based solutions
- Developed ability to explain business processes, accounting entries, and system configuration

13.3 Future Scope (Phase 2 Recommendations)

- Implement SAP SD (Order-to-Cash cycle) for NexGen's sales process
- Implement SAP PP (MRP) to automate Purchase Requisition generation from production planning
- Enable SAP SRM (Supplier Relationship Management) for vendor portal and e-sourcing
- Migrate to SAP S/4HANA for enhanced analytics and Fiori-based user interface
- Implement SAP Ariba for cloud-based procurement and supplier collaboration

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